ALAN TSENG

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SUMMARY

Embedded Software Engineer with 1+ years of experience in BMC firmware development at a leading ODM. Experienced in Yocto, OpenBMC, IPMI, and cloud-native infrastructure. Passionate about system software, low-level debugging, and integrating AI capabilities into embedded systems.

OBJECTIVE

Seeking full-time roles in embedded systems development, leveraging experience in firmware, cloud infrastructure, and cross-platform debugging.

EDUCATION

M.Sc., Computer Science, National Taiwan University — Aug 2021 - Feb 2024
Overall GPA: 4.12/4.3

EXPERIENCE

Embedded Systems Engineer – BMC Firmware (OpenBMC, Yocto) — Dec. 2024 - Present
Wiwynn, New Taipei City, Taiwan

- Developed a firmware validation tool based on a state machine, improving test coverage and reducing manual testing time by 30%.
- Leveraged GDB and gdbserver for cross-compilation debugging on ARM Cortex-A7, accelerating root cause analysis of kernel panics and peripheral failures.
- Enhanced OpenBMC features including IPMI command handlers, Redfish schema integration, and backend services using Yocto and BitBake.
- Collaborated with cross-functional teams on GB200-based L10 product development to deliver OCP-compliant BMC solutions.

Server Software Engineer – Provisioning Automation (Kubernetes) — Sep. 2024 - Dec. 2024
Pegatron, Taipei City, Taiwan

- Built a rack server testing system using Kubernetes, Helm, and Shell scripting to automate Windows/Linux OS deployments, reducing image size from 663 MB to 2.4 MB.

Software Developer Intern — Jul. 2022 - Nov. 2022
LINE Corporation, Taipei City, Taiwan

- Reduced API development time by 20% by introducing Swagger-driven codegen with Kubernetes deployment, DNS routing, and load balancing.
- Led 5 developers in building an NFT merchant platform using NextJS and Spring Boot during an internal hackathon.
- Presented Ethereum & dApp architecture in internal study sessions.
- Contributed to a scalable, data-driven B2B system with Nx monorepo practices.

SELECTED PROJECTS AND COURSES

Linux Kernel Internals Course

- GitHub: course_linux_kernel – Explored kernel memory management, process scheduling, and data structures in C.

Drum.io - Course Project

- Won top of the class with 50+ votes by developing a trainable air drum game using TensorFlow.js, Arduino, and WebSockets.

Adaptive Driving Beam System - Undergraduate Thesis

- Award a scholarship by the Ministry of Science and Technology, Taiwan, published in the 2020 IPPR Conference on Computer Vision, Graphics, and Image Processing (CVGIP) and Honored for top 1% research & graduation by solving the nighttime vehicle detection and designing an adaptive driving beam using CycleGAN, YOLO, Statistics and SVM.

HONORS

- **Research Excellence Award** – Honored for top 1% research and academic excellence upon graduation.
- **Intel Andy Grove Scholarship** – Selected from 2000+ global applicants for academic and leadership potential.
- **College Student Research Scholarship** – Awarded by Taiwan's Ministry of Science and Technology.

EXTRA-CURRICULAR ACTIVITIES

- **Lead, Developer Student Club (Google DSC)** – Led 70+ students, conducted sessions on ML, Python, Flutter, and Posenet.
- **Leader, ACM Competitive Programming Club** – Hosted weekly UVa and algorithmic coding sessions.
- **Volunteering**
  - Tech support at ART FUTURE blockchain exhibit.
  - Staff at SITCON and COSCUP for DSC and GDG events.

Skills

- **Core Expertise:** Embedded Software Development, Cloud-Native Applications, Web Systems
- **Programming Languages:** C/C++, Python, Shell, JavaScript, Golang
- **Cloud & Observability:** Kubernetes (server deployment), Docker, Grafana, Helm
- **Toolchain & Debugging:** Yocto Project, BitBake, Gnu Make, perf, valgrind, GDB
- **SoC Platforms:** ARM Cortex-A7, Cortex-A53, Cortex-M3
- **Hardware Interfaces:** I2C, SPI, GPIO, UART, ADC
- **AI & Data Science:** TensorFlow, Scikit-learn
- **Web Development:** Node.js, React.js